LocalCloudConfigurator

System Description

Abstract

This document describes the LocalCloudConfigurator support system of the Eclipse Arrowhead 5.2.2 framework, as implemented in `ah5-LocalCloudConfigurator-java` of the primary Java source tree (`Eclipse-Arrowhead-java`, Java-Spring with the vendored Eclipse Arrowhead common library) and deployed with the standalone Compose definition of that folder.

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1 Overview

This document describes the LocalCloudConfigurator system. LocalCloudConfigurator designs and fills one existing, provisioned Java Arrowhead local cloud: from a trusted component catalog it derives dependency-ordered plans, keeps saved designs, and executes reviewed plans through a host worker that checks every step again.

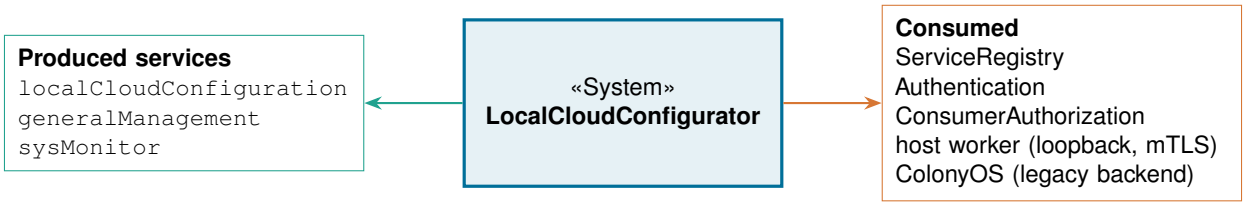


Figure 1: SysML-style block view of LocalCloudConfigurator with its produced and consumed services.

The rest of this document is organized as follows. In Section 1.1, we reference major prior art capabilities of the system. In Section 1.2, we describe the intended usage of the system. In Section 1.3, we describe fundamental properties provided by the system. In Section 1.4, we describe delimitations of capabilities of the system. In Section 2, we describe the abstract service functions consumed or produced by the system. In Section 3, we describe the security capabilities of the system.

1.1 Significant Prior Art

It ports the approach of the primary Go LocalCloudConfigurator (catalog, planner, reviewed plans, independent executor checks). Until November 2026 the Java version was a plain JDK HTTP server on port 8097 with file state, a shared operator token and a host worker on 8098; it was not an Arrowhead system.

1.2 How This System Is Meant to Be Used

The operator works in the graphical editor of LocalCloudManager (view *LC / Local cloud configuration*). LocalCloudManager consumes `localCloudConfiguration` through orchestration with its own certificate and identity token: it reads the catalog and the live status, saves designs, asks for a plan and applies the reviewed plan by its hash. LocalCloudConfigurator rebuilds the plan against the current runtime, and the host worker rebuilds and checks it once more before it runs the catalogued Compose actions.

1.3 System functionalities and properties

1.3.1 Functional properties of the system

- Trusted component catalog (`catalogs/javacloud.json`): components with their Compose services, owned databases, dependencies, provisioning and policy scripts, protected components and the default core.
- Plans: the default core plus the selected components and their dependencies, in dependency order (provision, start, policy), explicit stops in reverse order and an optional non-mutating verification; running services are not recreated unless `reapply` is set.
- Reviewed deployments: a plan is applied only if its hash still matches a freshly built plan; one deployment at a time; step results reported per step; interrupted deployments are marked failed and never replayed.
- Saved designs (selections and graphical layout) with revision checks; saving never changes the runtime.
- Execution backends: the host worker over mutual TLS (direct, default) or the retained ColonyOS workflows (legacy).
- `generalManagement` (logs, configuration) and native sysMonitor observations.

1.3.2 Configuration of system properties

The system is a Spring Boot application; `config/application.properties` is the deployment configuration (the packaged `application.properties` is a development sample). The properties that shape its behaviour are:

Property	Deployment value	Meaning
`server.port`	8492	Arrowhead connector (mutual TLS).
`server.ssl.*`	PKCS12 key store and truststore, `client-auth=need`	System certificate `LocalCloudConfigurator.<Cloud>.<Organization>.arrowhead.eu`.
`authentication.policy`	`outsourced`	Identity tokens of Authentication; own login with secret `ah.identity-password`.
`service.interface.policy`	`CERT_AUTH`	Policy of the registered interfaces.
`management.policy`	`sysop-only`	`generalManagement`.

Property	Deployment value	Meaning
`service.authorization.enabled`	true	ConsumerAuthorization check of every operation.
`lcc.api.whitelist`	empty	Systems that need no grant.
`arrowhead.ssl.crl-file`	`/crl/ca.crl`	CRL of the cloud CA (also for the worker connection).
`spring.datasource.*`	`localcloud configurator-db: 3306/ah_localcloud configurator, sslMode=verify-full`	Private MySQL schema; secret `ah.db-password`; `ddl-auto=validate`.
`lcc.catalog.file`	`/app/catalogs/ javacloud.json`	Component catalog.
`lcc.backend`	`direct`	`direct`: host worker; `colony`: ColonyOS (`COLONIES_*` variables).
`lcc.worker.url`	`https://host. docker.internal: 8493`	Host worker; only `127.0.0.1`, `localhost` or `host.docker. internal` with `https`.
`lcc.worker.server-name`	`LocalCloud ConfiguratorWorker`	Accepted system name of the worker certificate.
`sysmonitor.*`	enabled, `amqp-broker:5671`	Common sysMonitor publisher.
`arrowhead.security. profile`	`secure` (default)	The security profile guard refuses to start with TLS off, declared identities or the authorization check off; `development` only explicitly.

The host worker has its own properties (`lcc.worker.*`), described in the SysDD.

1.3.3 Data stored by the system

Private MySQL schema `ah_localcloudconfigurator`: saved designs, submitted deployments with the last reported state and step results, the active-deployment record and the log table. The host worker keeps its own file state (per-step results, its active-deployment marker) under `~/.local/share/JavaLocalCloudConfigurator/`.

1.3.4 Non functional properties

Deployments are serialized (one at a time). Status reads are cached for 2 s for display; plan and apply always read the runtime afresh. Interruption is not a transactional rollback: completed steps remain applied.

1.3.5 Stateful or stateless

Stateful (designs and deployment records in MySQL; worker file state).

1.4 Important Delimitations

The target is an existing, provisioned Java cloud: components are limited to the images, services and scripts of the catalog; arbitrary images, commands or secrets are not accepted. Starts use `--no-deps --no-build --pull never`. The default core, its infrastructure and protected components cannot be stopped. The catalog and its scripts are trusted host-administrator inputs.

2 Services

This section describes the produced and consumed services. Each produced service is specified by an SD document and one IDD document per interface.

2.1 Produced services

Service	Version	Interfaces	SD	IDD
`localCloudConfiguration`	1.0.0	HTTP(S)	[1]	[2]
`generalManagement`	1.0.0	HTTP(S)	–	–
`sysMonitor`	1.0.0	AMQP(S)	[3]	[4]

All HTTP services are registered on port 8492 with `generic_https` and policy `CERT_AUTH`. `generalManagement` (base path `/localcloudconfigurator/general/mgmt`) offers `get-log` (`POST /logs`) and `get-config` (`GET /get-config`) for Sysop. The consumer of `localCloudConfiguration` is LocalCloudManager [5].

2.2 Consumed services

Provider	Service	Operations	Use	Condition
ServiceRegistry	systemDiscovery, serviceDiscovery	register, revoke	Registration (common listener).	always
Authentication	identity	identity-login, identity-verify, identity-logout	Own token; verification of caller tokens.	always
Consumer Authorization	authorization	verify	Grant of each caller and operation (cached 30 s, fail closed).	always
Host worker	worker API (not an Arrowhead service)	status, deployments	Runtime status and execution of reviewed plans.	`lcc.backend=direct`
ColonyOS	Colonies RPC	submit, get process, workflows	Legacy execution backend.	`lcc.backend=colony`

3 Security

LocalCloudConfigurator is a Java-Spring support system. Its only connector (8492) uses HTTPS with mutual TLS and the X.509 certificate following the Arrowhead certificate profile (`LocalCloudConfigurator.<Cloud>.<Organization>.arrowhead.eu`).

3.1 Security model

- **Protocols:** HTTPS (TLS 1.3 and 1.2) with mutual TLS on 8492 and to the host worker; AMQP over TLS for `sysMonitor`; TLS to its database (`verify-full`).
- **Security profile:** the common security profile guard refuses insecure settings unless the `development` profile is selected explicitly.
- **Callers:** client certificate and identity token of the same system; every `localCloudConfiguration` operation needs Sysop, `lcc.api.whitelist` or a ConsumerAuthorization grant of that operation (LocalCloudManager); `generalManagement` is Sysop only. There is no shared operator token.
- **Host worker:** mutual TLS on host loopback only; the configurator presents its LocalCloudConfigurator certificate and accepts only a worker certificate named `LocalCloudConfiguratorWorker`; the worker accepts only clients named `LocalCloudConfigurator`; both check the chain against the cloud truststore and, when configured, the CRL.
- **Execution:** the worker trusts only its own catalog; it rebuilds the reviewed plan, checks claimed ownership, step kind, script and catalog hash, runs argument lists (no shell interpolation), bounds and redacts command output.
- **Data protection:** private schema with a DML-only account; secrets as Compose file secrets; read-only root filesystem, UID 10001.

For the Arrowhead certificate profile see github.com/eclipse-arrowhead/documentation.

4 References

[1] J. Delsing, “localCloudConfiguration – Service Description (LocalCloudConfigurator),” Eclipse Arrowhead, SD 5.2.2, 2026, localCloudConfigurator-v5.2.2-localCloudConfiguration-SD.pdf.

[2] ——, “localCloudConfiguration HTTP/TLS/JSON – Interface Design Description (LocalCloudConfigurator),” Eclipse Arrowhead, IDD 5.2.2, 2026, localCloudConfigurator-v5.2.2-localCloudConfiguration-HTTP-IDD.pdf.

[3] ——, “sysMonitor – Service Description (LocalCloudConfigurator),” Eclipse Arrowhead, SD 5.2.2, 2026, localCloudConfigurator-v5.2.2-sysMonitor-SD.pdf.

[4] ——, “sysMonitor AMQP/TLS/JSON – Interface Design Description (LocalCloudConfigurator),” Eclipse Arrowhead, IDD 5.2.2, 2026, localCloudConfigurator-v5.2.2-sysMonitor-AMQP-IDD.pdf.

[5] ——, “LocalCloudManager – System Description,” Eclipse Arrowhead, SysD 5.2.2, 2026, ah5-LocalCloudManager-java-spring/documentation/v5.2.2/LocalCloudManager-v5.2.2-SysD.pdf.

5 Revision History

5.1 Amendments

No.	Date	Version	Subject of Amendments	Author
1	2026-11-09	5.2.2	Document created for the Java-Spring LocalCloudConfigurator: Arrowhead support system on 8492, private MySQL schema, host worker on loopback 8493 with mutual TLS, direct and ColonyOS backends, LocalCloudManager as consumer.	Claude Opus 5.5 (draft)

5.2 Quality Assurance

No.	Date	Version	Approved by
1		5.2.2	